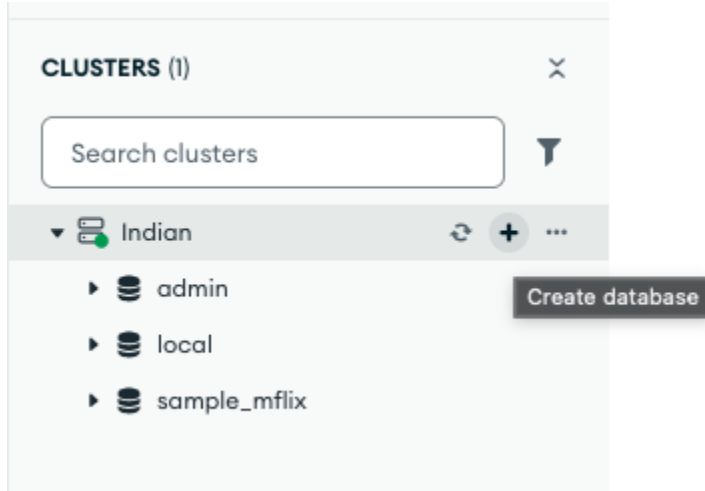


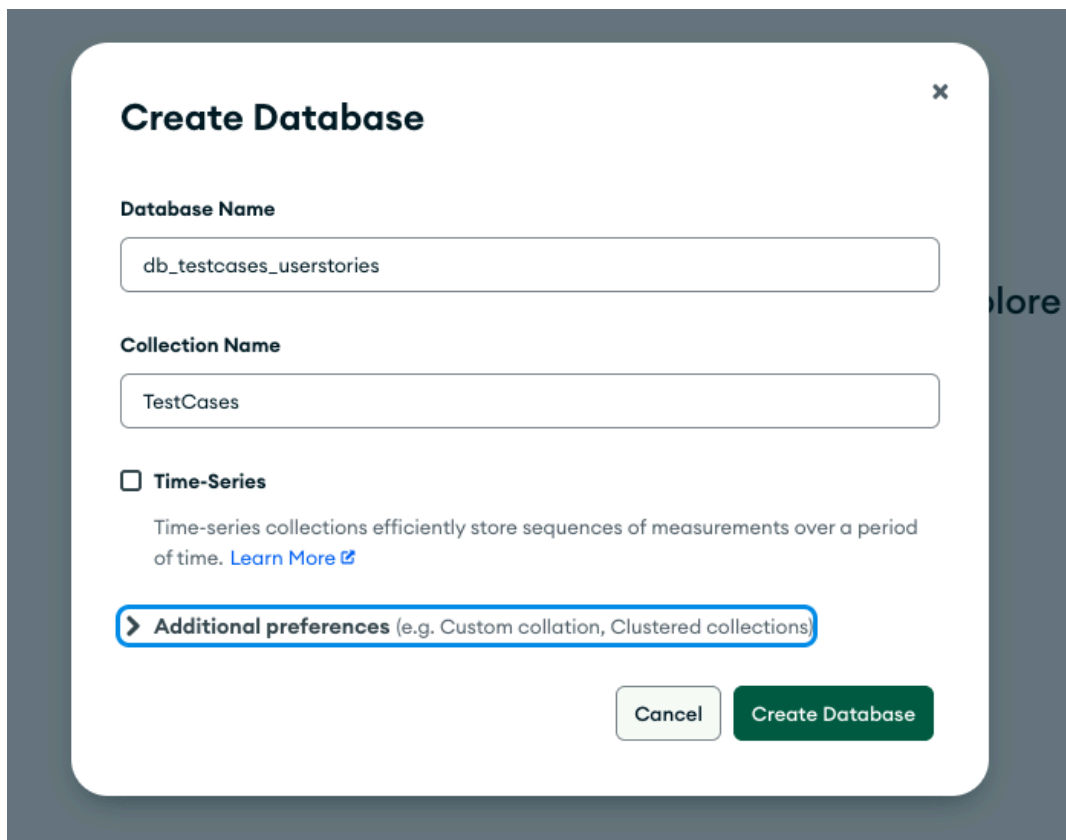
Assignment: Practice RAG ingestion frontend and backend

MongoDB Setup

1. Open mongoDB UI, Navigate to Data explorer and select the cluster (Indian)



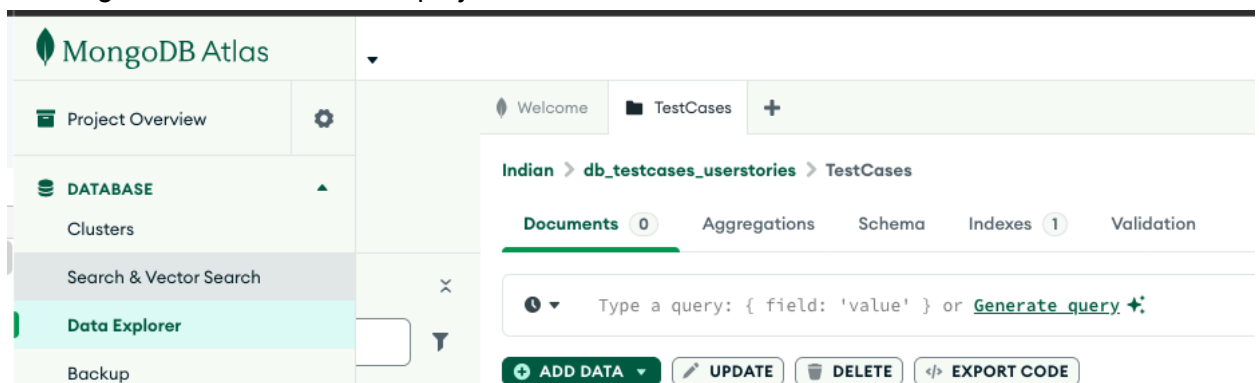
2. Click on create Database option ('+').
3. Provide a DB name (db_testcases_userstories) and Collection name (TestCases).

A screenshot of the 'Create Database' dialog box in MongoDB. The dialog has a title bar with a close button. Inside, there are two input fields: 'Database Name' with the value 'db_testcases_userstories' and 'Collection Name' with the value 'TestCases'. Below these fields is a checkbox labeled 'Time-Series' which is unchecked. Underneath the checkbox is a description: 'Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)'. At the bottom, there is a section titled 'Additional preferences (e.g. Custom collation, Clustered collections)' which is expanded. At the very bottom of the dialog are two buttons: 'Cancel' and 'Create Database'.

- Click 'Create Database'. DB should be created along with the collection



- Creating vector search -> From project overview select 'Search & Vector search'



- Click on 'Create search index' and provide below details
Type: Vector Search
How do you want to set up your vector data?: Bring your own embeddings
Index Name: vector_index_TestCase
Database and Collection: select the DB and collection created above.
Configuration Method: JSON Editor
- Click on Next. In the JSON editor paste the content from "testcases-vector-index.json" file in the rag-mongo-demo-ui repo. And click Next
- Click 'Create Vector search Index'
- Follow the same steps to create another Collection 'UserStories' and vector search 'vector_index_UserStories'.

Pro free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.

Database	Collection	Index Name	Status	Queryable	Type
db_testcases_userstories	TestCases	vector_index_TestCase	READY View status details	✓	vectorSearch
db_testcases_userstories	UserStories	vector_index_UserStories	READY View status details	✓	vectorSearch

Importing test cases from backend

1. Open the Repo rag-mongo-demo-ui in the vs editor.
2. Open .env file and configure as below

```
COLLECTION_NAME="TestCases"
VECTOR_INDEX_NAME="vector_index_TestCases"
BM25_INDEX_NAME="bm25_search"
USER_STORIES_COLLECTION_NAME="UserStories"
USER_STORIES_VECTOR_INDEX_NAME="vector_index_UserStories"
```
3. Open terminal for the repo in Vs studio and run 'npm install'
4. Then run 'node src/scripts/data-conversion/excel-to-json.js'

```
3053 node src/scripts/utilities/delete-all-documents.js
> node src/scripts/data-conversion/excel-to-json.js
✓ Converted 6354 rows from "testcases" into src/data/testcases.json
~/De/g/Week 7/Learning material/Day 2/rag-mongo-demo-ui main 78
```

5. Run 'node src/scripts/data-conversion/excel-to-json.js' to upload the test cases to the DB

```
3053 node src/scripts/utilities/delete-all-documents.js
> node src/scripts/data-conversion/excel-to-json.js
✓ Converted 6354 rows from "testcases" into src/data/testcases.json
> node src/scripts/embeddings/create-embeddings-batch-mistral.js
Input file: src/data/testcases.json
MISTRAL AI BATCH PROCESSING: 6354 test cases
Using Mistral AI Embedding API for Cost-Effective Processing!
Configuration:
Batch Size: 50 testcases per API call
Concurrent Batch Calls: 3
MongoDB Batch Size: 100
Delay Between Batch Groups: 1000ms
API: Mistral AI (mistral-embed)
Embedding Dimensions: 1024
Database: RagTestCases
Collection: TestCases
Total Batches: 128
Batch Groups: 43
Estimated Time: 2.9 minutes

[Batch 1/128] Processing 50 testcases with Mistral AI...
[Batch 2/128] Processing 50 testcases with Mistral AI...
[Batch 3/128] Processing 50 testcases with Mistral AI...
✓ [Batch 2/128] Success! Cost: $0.000818 | Tokens: 8175
```

6. Test cases are now available in the DB - Verified in UI

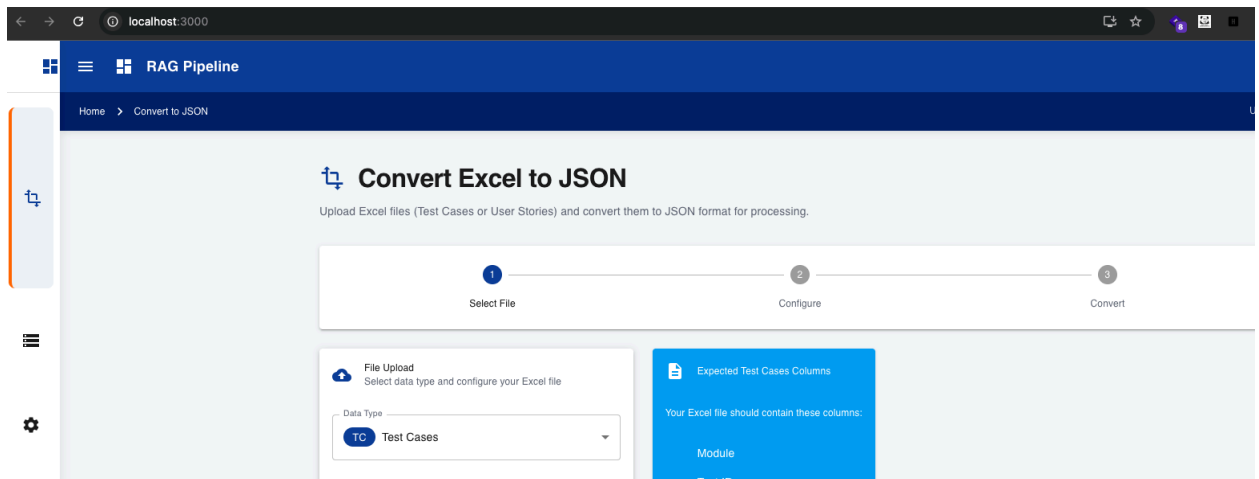
Type a query: (field: 'value') or [Generate query](#)

[ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT CODE](#) 25 1 ~ 25 of 6354

```
{
  "_id": ObjectId('6a1aaf95618e4410c2af2aac'),
  "module": "Registration",
  "id": "TC_005",
  "preRequisites": "PRN/ERN Number",
  "title": "Conversion from PRN/ERN Reg to UHID",
  "description": "Verify Conversion from PRN/ERN Reg to UHID",
  "steps": "1. Launch the application\n2. Login to the application with valid creden..",
  "expectedResults": "Should get convert successfully from PRN/ERN Reg to UHID",
  "automationManual": "Automated",
  "priority": "P1",
  "createdBy": "Prabakaran",
  "createdAt": 45165,
  "lastModifiedDate": 45176,
  "risk": "Low",
  "version": "v7.0",
  "type": "Integration",
  "linkedStories": "HC-080",
  "embedding": Array (1024),
  "createdAt": 2026-05-30T09:36:21.319+00:00
}
```

Importing User stories from frontend

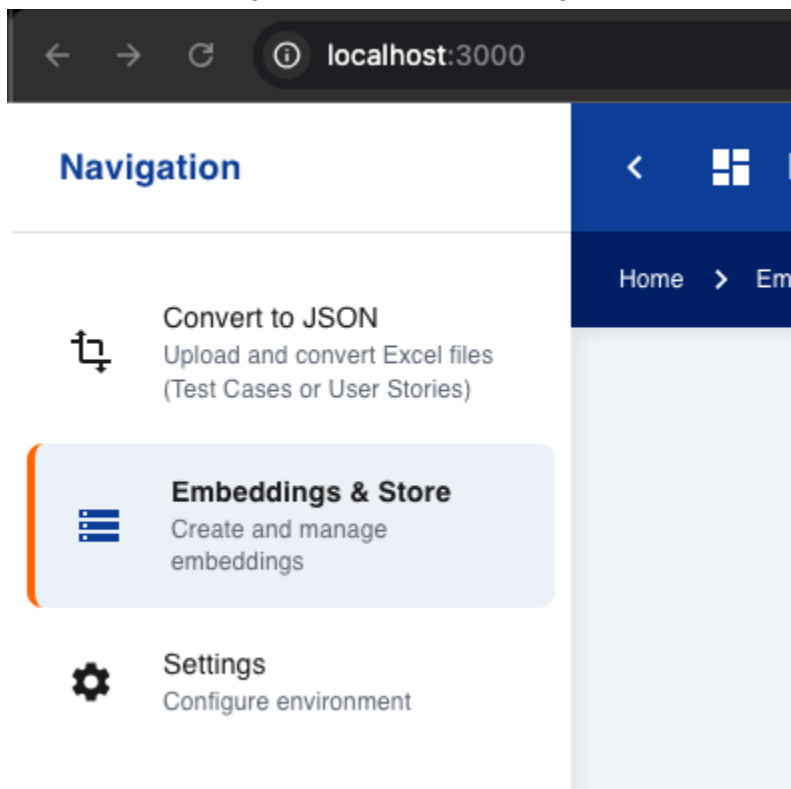
1. In the same setup as above, in the terminal run “npm run dev”
2. An UI will open with url “http://localhost:3000/”



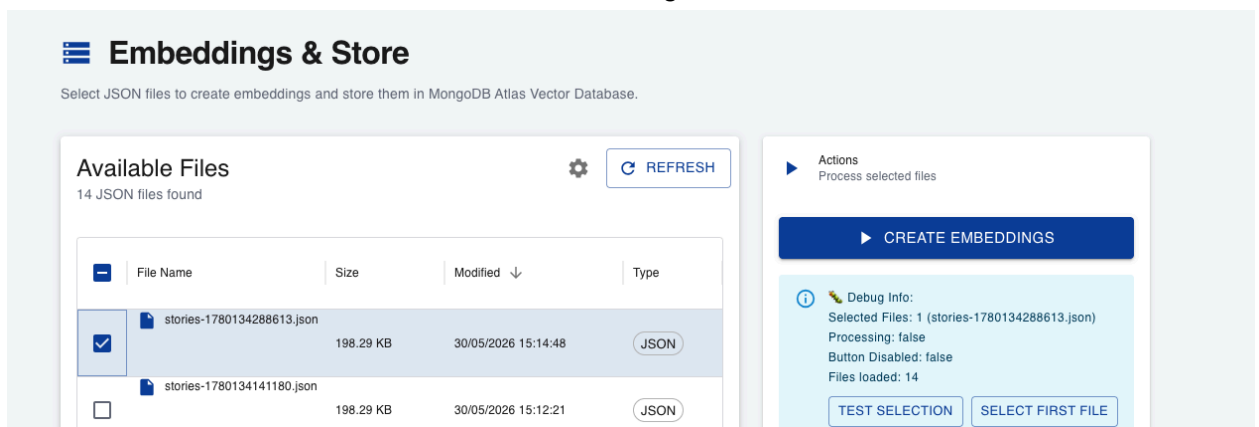
3. Select Data Type as user stories and select the user story (currently available in “rag-mongo-demo-ui/src/data/userstories.xlsx”)

A detailed view of the 'File Upload' configuration form. It includes a 'Data Type' dropdown menu with 'US User Stories' selected. Below this is a dashed box with a cloud icon and the text 'SELECT EXCEL FILE (USER STORIES)'. A file named 'userstories.xlsx' (25.6 KB) is shown as 'Selected'. There is a 'Sheet Name' input field containing the text 'stories', with a note below it: 'Name of the Excel sheet to convert (default: stories)'. At the bottom, there are two buttons: 'CONVERT USER STORIES' and 'RESET'.

- Click on Convert User Stories.
- The Converted user stories should be available at 'rag-mongo-demo-ui/src/data/'
- Click on Embeddings and Store in the navigation pane



- Sort by 'Modified' to get the latest file at top.
- Select the latest file and click on 'Create Embeddings'.



9. Once processed look into the UserStories collection having the user stories added.

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'TestCases' and 'UserStories'. Below the tabs, the breadcrumb path is 'Indian > db_testcases_userstories > UserStories'. On the right, there are buttons for 'View monitoring' and a status indicator. Below the breadcrumb, there are tabs for 'Documents' (190), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. A search bar contains the text 'Type a query: { field: 'value' } or [Generate query](#)'. To the right of the search bar are buttons for 'Explain', 'Reset', and 'Find'. Below the search bar, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'. To the right of these buttons is a dropdown menu set to '25' and a text '1 - 25 of 190'. Below the buttons, there is a document viewer showing a single document. The document has the following fields: '_id: ObjectId('6a1b1da51b07eab803091a3')', 'key: "HC-257"', 'summary: "Consultant Dashboard My Inpatients"', 'description: "As a user, I want to consultant dashboard my inpatients so that the re..."', 'status: Object', 'priority: Object', 'assignee: null', 'reporter: null', 'created: "2022-01-06T00:00:00.000Z"', 'updated: "2026-05-30T09:44:48.895Z"', 'components: Array (empty)', 'labels: Array (empty)', 'fixVersions: Array (empty)', 'storyPoints: null', 'project: "Health Care"', 'epic: "Emergency Management"', 'acceptanceCriteria: "Given a user with valid access, when they open the dashboard, then re..."', 'businessValue: ""', and 'risk: "Critical"'. There is a small 'Array (empty)' tooltip over the 'fixVersions' field.

```
{
  "_id": ObjectId('6a1b1da51b07eab803091a3'),
  "key": "HC-257",
  "summary": "Consultant Dashboard My Inpatients",
  "description": "As a user, I want to consultant dashboard my inpatients so that the re...",
  "status": Object,
  "priority": Object,
  "assignee": null,
  "reporter": null,
  "created": "2022-01-06T00:00:00.000Z",
  "updated": "2026-05-30T09:44:48.895Z",
  "components": Array (empty),
  "labels": Array (empty),
  "fixVersions": Array (empty),
  "storyPoints": null,
  "project": "Health Care",
  "epic": "Emergency Management",
  "acceptanceCriteria": "Given a user with valid access, when they open the dashboard, then re...",
  "businessValue": "",
  "risk": "Critical"
}
```